

A Technical Introduction to Bagging ensemble

Section 1

Example: ozone data To illustrate the basic principles of bagging, below is an analysis on the relationship between ozone and temperature (data from Rousseeuw and Leroy (1986), analysis done in R). The relationship between temperature and ozone appears to be nonlinear in this dataset, based on the scatter plot. To mathematically describe this relationship, LOESS smoothers (with bandwidth 0.5) are used. Rather than building a single smoother for the complete dataset, 100 bootstrap samples were drawn. Each sample is composed of a random subset of the original data and maintains a semblance of the master set's distribution and variability. For each bootstrap sample, a LOESS smoother was fit. Predictions from these 100 smoothers were then made across the range of the data. The black lines represent these initial predictions. The lines lack agreement in their predictions and tend to overfit their data points: evident by the wobbly flow of the lines.

Section 2

This process is repeated recursively for successive levels of the tree until the desired depth is reached. At the very bottom of the tree, samples that test positive for the final feature are generally classified as positive, while those that lack the feature are classified as negative. These trees are then used as predictors to classify new data.

Section 3

Creating the out-of-bag dataset The out-of-bag dataset represents the remaining people who were not in the bootstrap dataset. It can be calculated by taking the difference between the original and the bootstrap datasets. In this case, the remaining samples who were not selected are Emily, Jessie, George, Rachel, and Jamal. Keep in mind that since both datasets are sets, when taking the difference the duplicate names are ignored in the bootstrap dataset. The illustration below shows how the math is done:

Section 4

Creating the bootstrap dataset The bootstrap dataset is made by randomly picking objects from the original dataset. Also, it must be the same size as the original dataset. However, the difference is that the bootstrap dataset can have duplicate objects. Here is a simple example to demonstrate how it works along with the illustration below:

Section 5

Improving Random Forests and Bagging While the techniques described above utilize random forests and bagging (otherwise known as bootstrapping), there are certain techniques that can be used in order to improve their execution and voting time, their prediction accuracy, and their overall performance. The following are key steps in creating an efficient random forest:

Section 6

Creation of Decision Trees The next step of the algorithm involves the generation of decision trees from the bootstrapped dataset. To achieve this, the process examines each gene/feature and determines for how many samples the feature's presence or absence yields a positive or negative result. This information is then used to compute a confusion matrix, which lists the true positives, false positives, true negatives, and false negatives of the feature when used as a classifier. These features are then ranked according to various classification metrics based on their confusion matrices. Some common metrics include estimate of positive correctness (calculated by subtracting false positives from true positives), measure of "goodness", and information gain. These features are then used to partition the samples into two sets: those that possess the top feature, and those that do not. The diagram below shows a decision tree of depth two being used to classify data. For example, a data point that exhibits Feature 1, but not Feature 2, will be given a "No". Another point that does not exhibit Feature 1, but does exhibit Feature 3, will be given a "Yes".

Section 7

By taking the average of 100 smoothers, each corresponding to a subset of the original dataset, we arrive at one bagged predictor (red line). The red line's flow is stable and does not overly conform to any data point(s).

Section 8

Key Terms There are three types of datasets in bootstrap aggregating. These are the original, bootstrap, and out-of-bag datasets. Each section below will explain how each dataset is made except for the original dataset. The original dataset is whatever information is given.

Section 9

For a weak learner with high bias, bagging will also carry high bias into its aggregate Loss of interpretability of a model. Can be computationally expensive depending on the dataset.

Section 10

History The concept of bootstrap aggregating is derived from the concept of bootstrapping which was developed by Bradley Efron. Bootstrap aggregating was proposed by Leo Breiman who also coined the abbreviated term "bagging"

(bootstrap aggregating). Breiman developed the concept of bagging in 1994 to improve classification by combining classifications of randomly generated training sets. He argued, "If perturbing the learning set can cause significant changes in the predictor constructed, then bagging can improve accuracy".

Section 11

Random Forests The next part of the algorithm involves introducing yet another element of variability amongst the bootstrapped trees. In addition to each tree only examining a bootstrapped set of samples, only a small but consistent number of unique features are considered when ranking them as classifiers. This means that each tree only knows about the data pertaining to a small constant number of features, and a variable number of samples that is less than or equal to that of the original dataset. Consequently, the trees are more likely to return a wider array of answers, derived from more diverse knowledge. This results in a random forest, which possesses numerous benefits over a single decision tree generated without randomness. In a random forest, each tree "votes" on whether or not to classify a sample as positive based on its features. The sample is then classified based on majority vote. An example of this is given in the diagram below, where the four trees in a random forest vote on whether or not a patient with mutations A, B, F, and G has cancer. Since three out of four trees vote yes, the patient is then classified as cancer positive.

Section 12

Description of the technique Given a standard training set D of size n , bagging generates m new training sets D_i , each of size n' , by sampling from D uniformly and with replacement. By sampling with replacement, some observations may be repeated in each D_i . If $n' = n$, then for large n the set D_i is expected to have the fraction $(1 - 1/e)$ (~63.2%) of the unique samples of D , the rest being duplicates. This kind of sample is known as a bootstrap sample. Sampling with replacement ensures each bootstrap is independent from its peers, as it does not depend on previous chosen samples when sampling. Then, m models are fitted using the above bootstrap samples and combined by averaging the output (for regression) or voting (for classification).